SIMATS ENGINEERING

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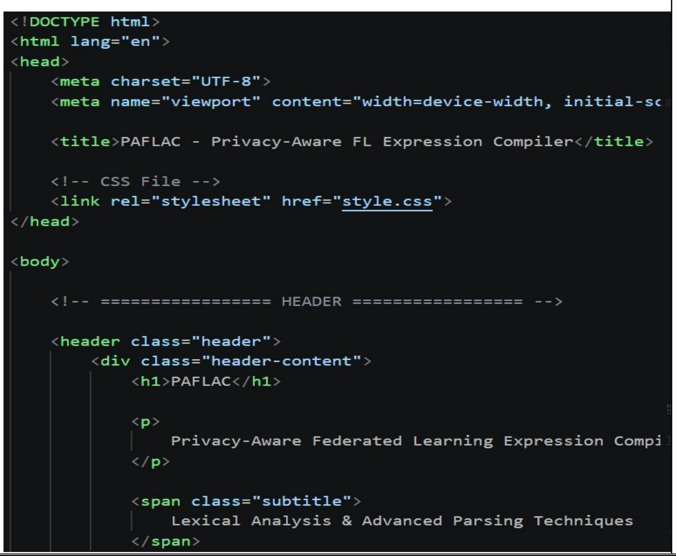
Course Code: CSA 1409 **Slot: C**

Course Name: COMPILER DESIGN

Course Faculty:Dr. G. ANITHA

Project Title: Privacy-Aware Federated Learning Expression Compiler

Module Photographs:



Project Description:

This project focuses on developing a Privacy-Aware Federated Learning Expression Compiler for secure Artificial Intelligence applications. In Federated Learning, sensitive data remains on client devices, while only model updates and privacy-related information are shared with the central aggregation system. The proposed compiler helps validate privacy-related expressions before they are used in the Federated Learning process.

The system performs lexical and syntax analysis of expressions containing privacy parameters such as privacy budget, noise strength, client trust, exposure risk, and security bonus. It identifies tokens, validates the grammar, checks operator precedence, and detects syntax errors. The project also uses parsing techniques such as LL(1) Predictive Parsing, Recursive Descent Parsing, Operator-Precedence Parsing, and SLR(1) Parsing.

The project can be extended to support advanced privacy-aware functions such as differential privacy, secure aggregation, noise addition, and privacy scoring. By comparing different parsing techniques, the project identifies suitable methods for processing dynamically generated privacy expressions. This helps improve the correctness, security, scalability, and reliability of privacy-aware Federated Learning systems.

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